Average value of $f(x, y, z) = x^2z + y^2z$ over a paraboloid solid

Problem

The average value of a function $f(x, y, z)$ over a solid region E is defined to be

$$f_{\text{ave}} \;=\; \frac{1}{V(E)} \iiint_E f(x, y, z)\, \mathrm{d}V,$$

where $V(E)$ is the volume of E. For instance, if ρ is a density function, then ρ_{ave} is the average density of E. Find the average value of the function

$$f(x, y, z) \;=\; x^2z + y^2z$$

over the region enclosed by the paraboloid $z = 1 - x^2 - y^2$ and the plane $z = 0$.

Notation used in this solution.

- E is the solid region, that is, the set of all points (x, y, z) inside the closed surface.
- $V(E) = \iiint_E 1\, \mathrm{d}V$ is the volume of E.
- $\mathrm{d}V$ is the volume element. In Cartesian coordinates $\mathrm{d}V = \mathrm{d}z\, \mathrm{d}y\, \mathrm{d}x$; in cylindrical coordinates $\mathrm{d}V = r\, \mathrm{d}z\, \mathrm{d}r\, \mathrm{d}\theta$.
- (r, θ, z) are cylindrical coordinates: $x = r\cos\theta$, $y = r\sin\theta$, $r \geq 0$, $0 \leq \theta < 2\pi$. In these coordinates $x^2 + y^2 = r^2$.
- The plane $x = 0$ is the set of points with x-coordinate 0. The plane $y = 0$ is defined in the same way.

Part 1: the average value

1. Description of the region

The two boundary surfaces are the paraboloid $z = 1 - x^2 - y^2$ and the plane $z = 0$. They meet where both equations hold at the same time:

$$x^2 + y^2 = 1, \qquad z = 0.$$

That intersection is the unit circle in the plane $z = 0$. The paraboloid is a cap that opens downward, with apex $(0, 0, 1)$. So the solid E is the set

$$E \;=\; \big\{(x, y, z) :\; x^2 + y^2 \leq 1,\; 0 \leq z \leq 1 - x^2 - y^2\big\}.$$

Figure 1 shows E: a flat disk as the floor and a curved paraboloid as the roof.

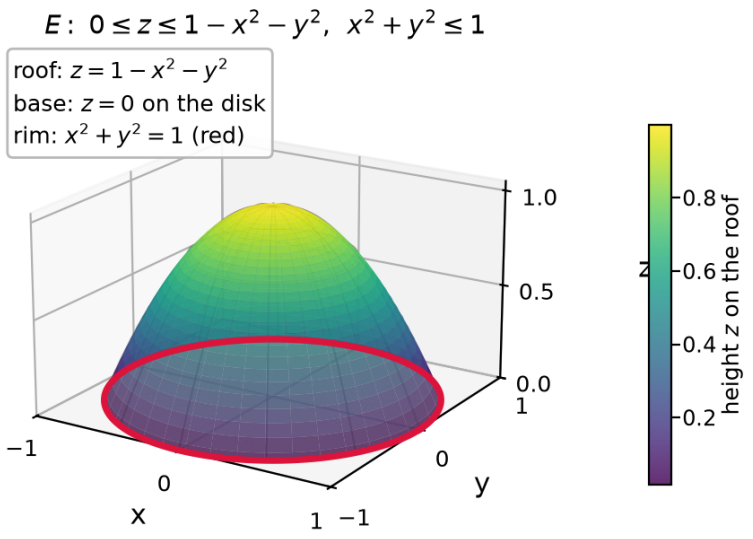


Figure 1: The solid E. The roof is the paraboloid $z = 1 - x^2 - y^2$. The floor is the disk $x^2 + y^2 \le 1$ in the plane $z = 0$. The red curve is the rim where the two surfaces meet. The color of the roof shows the height z.

2. The region is z-simple, so the integral is legal

For each point (x, y) in the unit disk $D = \{(x, y) : x^2 + y^2 \le 1\}$, the vertical line over (x, y) enters E at $z = 0$ and leaves E at $z = 1 - x^2 - y^2$. Both bounds are continuous functions of (x, y), and the integrand f is a polynomial, so it is continuous on E. By Fubini's theorem the triple integral equals an iterated integral in the order $\mathrm{d}z$, then $\mathrm{d}y\,\mathrm{d}x$ (or $\mathrm{d}z$, then $\mathrm{d}r\,\mathrm{d}\theta$):

$$\iiint_E f \,\mathrm{d}V \;=\; \iint_D \left(\int_0^{1-x^2-y^2} f(x, y, z)\,\mathrm{d}z \right) \mathrm{d}A.$$

Because f and the bounds depend on x and y only through $x^2 + y^2$, cylindrical coordinates are the efficient choice. With $x^2 + y^2 = r^2$ and $\mathrm{d}V = r\,\mathrm{d}z\,\mathrm{d}r\,\mathrm{d}\theta$, the region becomes the box

$$0 \le r \le 1, \qquad 0 \le \theta \le 2\pi, \qquad 0 \le z \le 1 - r^2,$$

and the order $\mathrm{d}z,\ \mathrm{d}r,\ \mathrm{d}\theta$ is legal for the same reason.

3. Why the symmetry does not make the integral vanish

This step needs care, because two different facts are easy to mix up.

Fact A: the region is symmetric. The solid E is symmetric about the plane $x = 0$ and about the plane $y = 0$. If $(x, y, z) \in E$, then $(-x, y, z) \in E$ and $(x, -y, z) \in E$.

Fact B: the integrand is even. Compute

$$f(-x, y, z) = (-x)^2 z + y^2 z = x^2 z + y^2 z = f(x, y, z),$$

and in the same way $f(x, -y, z) = f(x, y, z)$ and $f(-x, -y, z) = f(x, y, z)$.

An even function on a symmetric region *doubles* the contribution of each half; it does not cancel it. The integral of the half $x \geq 0$ equals the integral of the half $x \leq 0$, so

$$\iiint_E f \, dV \;=\; 4 \iiint_{E \cap \{x \geq 0,\, y \geq 0\}} f \, dV.$$

The factor 4 is the only effect of the symmetry on this problem.

Cancellation needs a different property, namely oddness. The test is $f(-x, y, z) = -f(x, y, z)$. Take $g(x, y, z) = xz$. Then

$$g(-x, y, z) = (-x)z = -xz = -g(x, y, z),$$

and the pair of points (x, y, z) and $(-x, y, z)$ contributes $xz + (-xz) = 0$. So $\iiint_E xz \, dV = 0$. The function in this exercise is not of that type: it contains x^2, not x, and x^2 stays positive when x changes sign.

The value stays positive. On E we have $z \geq 0$ and $x^2 + y^2 = r^2 \geq 0$, so $f = z(x^2 + y^2) \geq 0$. It is zero only on the floor $z = 0$ and on the rim $r = 1$, and it is positive inside E. Therefore the integral is strictly positive, and so is f_{ave}.

Figure 2 shows this in two pictures. The slice at $y = 0$ has the same height at x and at $-x$. The second panel shows the value of f on the solid. On the roof $z = 1 - r^2$, so $f = z(1 - z)$. That product is largest at $z = 1/2$, where $f = 1/4$. So the largest values of f lie on the circle $r^2 = 1/2$ at height $z = 1/2$. The value is zero on the floor and on the rim, and f is small near the apex.

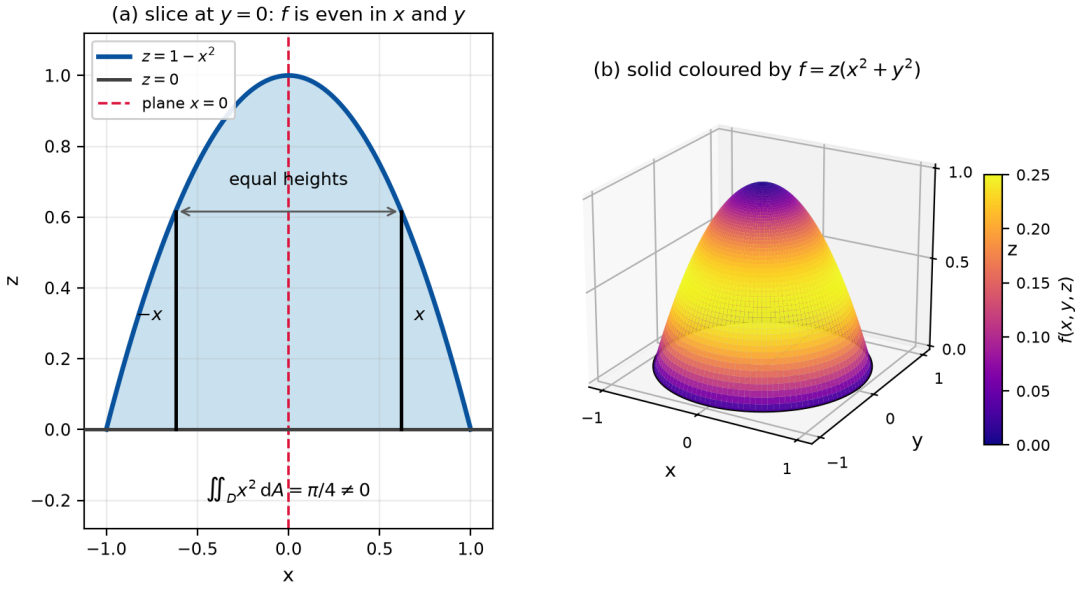


Figure 2: (a) The cross-section of E at $y = 0$. The two vertical segments have equal length, because the solid has the same height at x and at $-x$. This equality makes the integral larger by a factor 4; it does not make the integral zero. (b) The solid colored by the value of $f = z(x^2 + y^2)$. Dark purple means a small value and yellow means a large value. The value is zero on the whole floor $z = 0$ and on the rim $r = 1$.

4. The triple integral in cylindrical coordinates

On E we have $x^2+y^2=r^2$, so $f=r^2z$. The extra factor r in $\mathrm{d}V$ is the Jacobian of the cylindrical coordinates. So the integrand becomes r^2z and the volume element becomes $r\,\mathrm{d}z\,\mathrm{d}r\,\mathrm{d}\theta$:

$$\iiint_E f\,\mathrm{d}V \;=\; \int_0^{2\pi}\!\!\int_0^{1}\!\!\int_0^{1-r^2} r^2z\cdot r\,\mathrm{d}z\,\mathrm{d}r\,\mathrm{d}\theta \;=\; \left(\int_0^{2\pi}\mathrm{d}\theta\right)\left(\int_0^1 r^3\left[\frac{z^2}{2}\right]_0^{1-r^2}\mathrm{d}r\right).$$

The inner antiderivative is $\int z\,\mathrm{d}z = z^2/2$. Evaluate it at the two bounds:

$$\int_0^{2\pi}\!\!\int_0^1 r^3\cdot\frac{(1-r^2)^2}{2}\,\mathrm{d}r\,\mathrm{d}\theta \;=\; 2\pi\cdot\frac{1}{2}\int_0^1 r^3(1-2r^2+r^4)\,\mathrm{d}r.$$

Expand and integrate term by term, using $\int_0^1 r^n\,\mathrm{d}r = 1/(n+1)$:

$$\int_0^1\left(r^3-2r^5+r^7\right)\mathrm{d}r = \frac{1}{4}-\frac{2}{6}+\frac{1}{8} = \frac{6-8+3}{24} = \frac{1}{24}.$$

Therefore

$$\iiint_E\left(x^2z+y^2z\right)\mathrm{d}V \;=\; 2\pi\cdot\frac{1}{2}\cdot\frac{1}{24} \;=\; \frac{\pi}{24} \;\approx\; 0.1308997.$$

5. The volume $V(E)$

The same region with integrand 1 gives the volume:

$$V(E) = \int_0^{2\pi}\!\!\int_0^{1}\!\!\int_0^{1-r^2} r\,\mathrm{d}z\,\mathrm{d}r\,\mathrm{d}\theta = 2\pi\int_0^1 r(1-r^2)\,\mathrm{d}r$$
$$= 2\pi\left[\frac{r^2}{2}-\frac{r^4}{4}\right]_0^1 = 2\pi\left(\frac{1}{2}-\frac{1}{4}\right) = \frac{\pi}{2}\approx 1.5707963.$$

As an independent check, slice E horizontally at height z. The cross-section is the disk $x^2+y^2\le 1-z$ with area $\pi(1-z)$, so $V(E)=\int_0^1\pi(1-z)\,\mathrm{d}z=\pi/2$. Both routes agree.

6. The average value

Divide the integral by the volume:

$$f_{\text{ave}} \;=\; \frac{1}{V(E)}\iiint_E f\,\mathrm{d}V \;=\; \frac{\pi/24}{\pi/2} \;=\; \frac{\pi}{24}\cdot\frac{2}{\pi} \;=\; \frac{1}{12} \;\approx\; 0.0833333.$$

The result is a number, as it must be: an average value of a scalar function is a scalar.

7. Numerical confirmation

Table 1 lists the exact values and the independent numerical values. The scripts use Gauss–Legendre quadrature with 200 nodes in each direction, a Monte Carlo estimate with 2×10^6 random points, and horizontal slices. Every number agrees with the exact value to the stated tolerance.

The Monte Carlo estimates have statistical error. With 2×10^6 samples the standard error of the volume estimate is about 5×10^{-4}, and the observed errors are of that size.

Quantity	Exact value	Numerical value	Method
$\iiint_E f \, \mathrm{d}V$	$\pi/24 = 0.1308996939$	0.130899693900	Gauss–Legendre
$\iiint_E f \, \mathrm{d}V$	$\pi/24$	0.130826	Monte Carlo
$V(E)$	$\pi/2 = 1.5707963268$	1.571310	Monte Carlo
$V(E)$	$\pi/2$	1.570796326795	Gauss–Legendre
f_{ave}	$1/12 = 0.0833333333$	0.083333333333	quotient of the above

Table 1: Exact values from sympy and independent numerical values from numpy.

How to solve problems of this kind

1. **Write the region as inequalities.** Find where the boundary surfaces meet. Then decide which variable has one fixed pair of bounds. Here the vertical direction is the simple one: $0 \le z \le 1 - r^2$.

2. **Choose coordinates that match the symmetry.** A region described by $x^2 + y^2$ is a region for cylindrical coordinates, and then $\mathrm{d}V$ gains the factor r. A region described by $x^2 + y^2 + z^2$ is a region for spherical coordinates.

3. **Test the symmetry of the integrand, not only of the region.** Compare $f(-x, y, z)$ with $f(x, y, z)$. If the result is f, the integral is even and the symmetry gives a factor. If the result is $-f$, the integral over the symmetric region is zero. A symmetric region alone says nothing about the value.

4. **Check the sign.** If $f \ge 0$ on E and $f > 0$ on a set of positive volume, then the integral is positive. This single check catches most wrong cancellations.

5. **Compute the volume separately.** The average value needs $V(E)$. Use the same bounds with integrand 1, or use a second slicing direction, and compare the two results.

6. **Verify twice.** Do one exact computation and one independent numerical computation. If the two disagree, the error is in the setup, not in the arithmetic.

Answer

$$\boxed{f_{\text{ave}} \;=\; \frac{1}{V(E)} \iiint_E \left(x^2 z + y^2 z\right) \mathrm{d}V \;=\; \frac{\pi/24}{\pi/2} \;=\; \frac{1}{12} \;\approx\; 0.0833}$$

The symmetry of E about the planes $x = 0$ and $y = 0$ multiplies the integral by 4. It does not make the integral zero, because f is even in x and in y, and $f \ge 0$ on E.